CITY UNIVERSITY OF CAIRO
Faculty of Engineering
Basic Sciences Department
Course Name: **Mathematics 1**
Course Code: **BAS 111**
Date: **23 /03/2024** ***Makeup Midterm Exam***

Academic Year: **2024/2025**
Semester: **Spring**
Level: **Freshman**
Time: **60 min.**
Examiner: **Dr. Zaher Hassanein**
Total Marks**: 25 Marks**

Student Name: **Student ID:**

Remarks: - The Exam Consists of **TWO** Questions in **THREE** Pages

PART A: [12 Marks]

Find $\frac{dy}{dx}$ the first derivative of the following questions (**Choose 4 parts)**

i) $y = e^{3x} + x^e + x^3 + 3^x + 3$ **(3 Marks)**

ii) $y = \ln \sin 3x + \sinh^{-1}(x^3 - 1)$ **(3 Marks)**

iii) $y = (x^3 - 5)^9 (x^2 - 5x\sqrt{x} - 1)^5$ **(3 Marks)**

iv) $x^2 e^y + \sin \ln y - \tanh^{-1} x = 7$ **(3 Marks)**

v) $y = 3^{4x+1} \tan^{-1}(x + 2)$ **(3 Marks)**

vi) $f(x) = (x^2 + 1)^{\sec 2x}$ **(3 Marks)**

Question (2): [13 Marks]

(a) Evaluate the following integrals **(Choose 3 parts)**

i) $\int\left(x^6-\sqrt{x}+\frac{1}{x^4}+8\right)dx$ **(3 Marks)**

ii) $\int x^4 e^{x^5+5}\,dx$ **(3 Marks)**

iii) $\int\frac{x^2}{x^3+6}\,dx$ **(3 Marks)**

iv) $\int\frac{1}{x^2+9}\,dx$ **(3 Marks)**

(b) Calculate the integral $\int_0^{\pi/2} \frac{\sin 2x}{\sqrt{4+\cos^2 x}}\,dx$. **(4 Marks)**

GOOD LUCK

Committee of Examiners:1-Dr. Zaher Hassanein

2- Ass. Prof. Dr. Mohamed Abdel-Aziz